KARLOSH YADAV

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Education

Indian Institute of Science, Bangalore — 7.0/10
Master of Technology, Computer Science and Engineering — *Aug. 2025 – June 2027*

Visvesvaraya Technological University — 8.75/10
Bachelor of Engineering, Computer Science and Engineering — *Aug. 2020 – June 2024*

Projects

Benchmarking Deep Learning Models for Brain Tumor Classification: A Comparative Study of YOLOv8, ResNet50, and MobileNetV3 — Aug 2025 – Dec 2025

- Conducted a comparative analysis of state-of-the-art deep learning architectures for MRI-based brain tumor classification to evaluate performance across model families (YOLOv8, ResNet50, MobileNetV3).
- Performed systematic hyperparameter optimization and ablation studies to analyze factors influencing model accuracy and generalization.
- Applied K-fold cross-validation and rigorous evaluation to ensure statistical reliability and reproducibility.
- Developed a TensorFlow training and evaluation pipeline ensuring stable training and fair model comparison.

Heart Disease Prediction System (Ensemble Learning) — Sep 2023 – May 2024

- Built a prediction system using a Soft Voting ensemble of Logistic Regression, Random Forest, and XGBoost.
- Performed comprehensive data preprocessing including missing value handling, categorical encoding, feature scaling, and train-test splitting.
- Addressed class imbalance using SMOTE and optimized models using K-fold cross-validation and GridSearchCV.

Machine Learning Algorithms from Scratch

- Implemented supervised learning algorithms from scratch using NumPy, including Linear Regression, Logistic Regression, Decision Tree, and Random Forest without relying on high-level ML libraries.
- Derived and implemented mathematical formulations such as MSE loss, Binary Cross-Entropy, Gradient Descent optimization, and Information Gain for model training and splitting.
- Applied Random Forest with bootstrap sampling and feature randomization for improved generalization.

Vehicle Care Management System

- Designed a normalized relational database schema with ACID-compliant transaction management and implemented concurrency control to handle simultaneous service updates and prevent data inconsistencies.
- Implemented JWT-based stateless authentication, Role-Based Access Control (RBAC), and bcrypt-secured password management.
- Configured secure environment variable management and CORS policies to ensure safe cross-origin communication and credential handling in production.

Skills

Technical: C++, Python, MySQL, Machine Learning, Data Structures and Algorithms, Software Development Life Cycle, Database Management, Object-Oriented Programming (OOP), System Design (Basics)
Languages: English, Hindi, Maithili, Nepali

Certifications

Graduate Record Examination : 333/340
C++ for Everyone: University of California (2021)
Machine Learning for All: University of London (2023)

Extra-Curricular Activities

President: International Student Club (ISC) (04/2023 – 04/2024)
Secretary: Soft Research Computing Society (04/2022 – 04/2023)
Chess: Inter-Institution level player
Cricket: College and District Cricket Team.